

ASHUTOSH SHARMA

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AI Engineer — Generative AI / LLM Systems

PROFESSIONAL SUMMARY

AI Engineer with 3.5+ years shipping production RAG and agentic-LLM systems for enterprise document intelligence at T-Systems. Delivered 5+ end-to-end LLM products on LlamaIndex, LangGraph, FastAPI, and pgvector — owning retrieval, agent orchestration, evaluation (RAGAS), and async streaming.

TECHNICAL SKILLS

Generative AI & LLM: RAG (hybrid search, CRAG), LLM agents & ReAct, function calling, structured output, prompt engineering, embeddings, vector search, RAG evaluation (RAGAS), guardrails. **Frameworks:** LlamaIndex, LangChain, LangGraph, Hugging Face Transformers. **Models:** Llama 3, GPT, Claude.

Backend & Serving: Python (async), FastAPI, PostgreSQL + pgvector, SQLAlchemy (async), Celery, RabbitMQ, WebSocket / SSE streaming, REST APIs, Pydantic.

Cloud & MLOps: Docker, Kubernetes, Helm, CI/CD, Git, S3 / object storage, Phoenix tracing, Prometheus, Loguru.

ML Foundations: PyTorch, scikit-learn, NLP, classification, topic modeling (LDA).

PROFESSIONAL EXPERIENCE

Data Scientist — Generative AI Engineering | T-Systems ICT India Pvt. Ltd.

December 2022 – Present

AI Operator — Agentic RAG & Sandboxed Agent Platform *(May 2025 – Present)*

- Core engineer on a backend RAG + agent platform (FastAPI + LlamaIndex) serving Q&A over 10,000+ enterprise documents: hybrid retrieval on a custom pgvector store (semantic ANN + BM25), Cohere reranking, and streamed generation through an internal LLM gateway — cutting p95 response latency ~35% under production load.
- Architected the platform's sandboxed agent execution: per-session gVisor-isolated Kubernetes pods running LangGraph / deepagents, with a two-container design that keeps secrets out of the code workspace so model-generated code runs fully isolated from host and network.
- Partnered with the security team to design a 12-layer defense-in-depth security model (syscall filtering, network-egress control, path sandboxing, per-agent quotas, secret scrubbing) treating the LLM as untrusted — aligned to the OWASP LLM Top 10.
- Engineered multi-tool agentic workflows on LlamaIndex's Workflow API across 6 retrieval tools (query decomposition, parallel ReAct search, synthesis, citation / source filtering); raised RAGAS faithfulness from ~40% to 80%+ via reranking and eval-driven prompt tuning, with loop-prevention guards on the ReAct agent.
- Stood up a RAGAS + Phoenix evaluation pipeline running 1,000+ automated evals/week to tune prompts and retrieval parameters, and presented evaluation findings to engineering leadership.
- Extended ingestion from PDF-only to 6 formats across 15+ enterprise sources on Celery / RabbitMQ workers (per-child memory caps); served async streaming (astream_chat / SSE) at sub-second time-to-first-token, with Keycloak JWT auth and per-tenant isolation via service_key.
- Created a document-to-story capability (Story Generator): ephemeral PDF/DOCX parsing plus streaming chat that produces agile Epics and user stories, including auto-generated Epic-level Mermaid diagrams.

Insurance Healthcare Agent — Agentic RAG *(December 2024 – April 2025)*

- Built an agentic assistant that answered policy questions by combining document retrieval with structured PostgreSQL lookups (function-calling tools), cutting query-resolution time ~40% and average wait time ~70% with low-confidence fallback handling.
- Ran in production at 10,000+ monthly queries with 99%+ uptime and graceful degradation under load.

CrewAI Multi-Agent Content System *(July 2024 – November 2024)*

- Developed a multi-agent content pipeline (CrewAI) with specialized researcher, writer, and editor agents handing off through a shared workflow across 5 content types, cutting draft turnaround ~50%.

Topic Modeling for Document Clustering *(February 2024 – July 2024)*

- Clustered 1,000+ enterprise documents with LDA topic modeling, then trained a LightGBM classifier (~90% accuracy) on the extracted topics to automate categorization that had been manual.

Enterprise Data Analytics & ML *(December 2022 – January 2024)*

- Trained customer segmentation and classification models (XGBoost, Random Forest, ~87% accuracy) plus the ETL + feature-engineering pipelines feeding them over 50K+ records; delivered analytics dashboards for enterprise clients.

Intern Data Scientist | Analytics Labs

November 2021 – November 2022

- Built a Random Forest model for customer-attribution analysis (~12% reduction in targeted-segment churn) and defined KPIs for real-time tracking across 500+ devices.

EDUCATION

M.Sc. Data Science (part-time) | Manipal Academy of Higher Education | July 2022 – July 2024

B.Sc. Computer Science, Mathematics & Statistics | Bangalore North University | July 2018 – November 2021

CERTIFICATIONS

- Generative AI with Large Language Models — DeepLearning.AI / Coursera
- LangChain for LLM Application Development — DeepLearning.AI
- Machine Learning Specialization — Coursera (Andrew Ng)

LANGUAGES

English — Professional Working Proficiency | Hindi — Native